

SAO JOSE DOS CAMPOS, Brazil

WMO: 838093

Lat: 23.229S

Lon: 45.862W

Elev: 2120

StdP: 13.60

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	47.9	50.1	39.4	38.4	70.1	42.9	44.0	67.5	16.7	68.5	14.9	68.6	1.4	180	0.291

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.5	93.2	69.3	91.2	69.6	89.4	69.8	75.2	84.1	74.2	83.0	73.4	82.0	6.8	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.4	134.5	77.0	71.9	127.7	76.0	71.4	125.6	75.7	40.3	83.9	39.3	83.7	38.4	81.3	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max					
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.4	15.6	12.9	DB	42.7	97.0	4.4	2.7	39.5	98.9	36.9	100.5	34.4	102.0	31.2	104.0	
			WB	41.7	78.1	4.3	3.0	38.6	80.3	36.1	82.0	33.6	83.7	30.5	85.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.3	76.6	77.5	75.9	72.1	67.2	64.7	64.4	66.8	69.7	72.2	73.0	75.5
	DBStd	6.25	3.89	3.27	3.21	4.09	4.19	4.38	4.87	4.79	5.39	5.17	4.62	3.88
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	217	0	0	0	2	26	55	69	38	18	6	3	0
	CDD50	7764	825	770	803	664	534	442	445	521	591	689	689	791
	CDD65	2507	360	350	338	216	95	47	49	94	159	231	242	326
	CDH74	16823	2340	2405	2043	1230	502	341	464	914	1319	1603	1530	2132
	CDH80	5963	929	972	710	341	83	49	73	277	542	647	540	800
Wind	WSAvg	4.4	4.3	4.2	4.1	4.0	3.7	3.2	3.4	4.2	5.4	5.7	5.7	5.1
Precipitation	PrecAvg	54.1	9.3	8.2	7.2	2.6	2.2	1.3	1.3	1.1	3.3	4.9	5.6	6.8
	PrecMax	71.2	15.2	17.7	13.4	6.1	4.2	5.5	3.7	2.5	6.5	9.6	9.7	10.3
	PrecMin	32.0	3.3	2.7	3.6	0.6	0.3	0.0	0.2	0.1	0.2	1.3	2.6	3.8
	PrecStd	9.3	3.5	3.5	2.2	1.3	1.1	1.3	1.1	0.7	1.8	2.0	1.9	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	96.4	94.6	91.5	89.5	84.6	84.4	84.6	90.0	95.1	95.4	93.3	93.4
		MCWB	71.7	70.1	71.9	69.7	67.4	65.3	63.5	63.4	65.4	67.7	69.8	69.8
	2%	DB	91.5	91.4	89.4	86.4	82.4	80.8	82.2	86.1	90.0	91.3	89.4	91.0
		MCWB	71.5	71.1	71.0	69.6	66.8	64.5	62.6	62.7	65.0	67.7	69.7	69.8
	5%	DB	88.2	89.4	87.0	84.2	79.2	78.4	79.1	82.8	86.2	87.6	86.1	87.9
		MCWB	71.6	71.5	71.1	69.5	65.9	63.8	62.5	62.7	64.5	67.6	69.2	69.9
	10%	DB	85.7	86.2	84.2	81.0	76.7	75.1	76.5	79.1	82.2	82.7	82.6	84.5
		MCWB	71.7	71.5	71.3	68.4	65.2	63.2	62.0	62.2	64.7	67.5	68.7	70.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	76.9	76.6	75.9	73.8	71.1	68.7	67.1	68.3	71.1	73.2	74.2	75.1
		MCDB	83.4	86.2	83.4	82.8	79.0	76.9	76.9	80.1	83.0	83.5	83.2	82.8
	2%	WB	75.2	74.9	74.4	72.5	69.2	66.6	65.5	65.9	69.2	71.4	72.6	73.6
		MCDB	84.2	84.5	82.6	80.9	76.8	75.1	75.0	78.1	79.9	82.9	82.1	82.8
	5%	WB	74.0	73.8	73.4	71.3	67.9	65.4	64.2	64.6	67.7	70.1	71.3	72.6
		MCDB	82.6	82.5	81.6	78.8	75.4	73.9	73.9	76.6	78.3	80.4	80.5	81.7
	10%	WB	73.0	72.8	72.5	70.2	66.4	64.1	63.0	63.4	66.3	68.9	70.3	71.6
		MCDB	81.1	81.0	80.1	77.1	73.4	72.2	72.4	74.4	76.6	78.1	78.6	80.3
Mean Daily Temperature Range	5% DB	MDBR	18.0	18.5	17.5	17.9	18.3	20.0	21.7	23.1	21.1	19.4	18.7	18.6
		MCDBR	22.4	22.5	20.8	21.0	22.2	24.7	27.1	29.7	29.8	27.2	24.0	22.8
	5% WB	MCWBR	7.3	6.8	6.7	7.2	8.9	10.3	10.5	10.0	9.3	8.4	8.0	7.4
		MCWBR	19.3	19.6	18.4	18.2	19.8	20.4	22.4	24.4	23.1	21.7	20.6	19.4
Clear-Sky Solar Irradiance	taub	0.396	0.393	0.385	0.364	0.352	0.346	0.356	0.391	0.482	0.455	0.430	0.404	0.404
		2.463	2.452	2.477	2.508	2.509	2.512	2.465	2.340	2.073	2.218	2.305	2.415	2.415
	Ebn at Noon	301	298	291	284	274	270	270	271	258	276	289	299	299
	Edh at Noon	38	38	35	31	29	28	30	37	52	47	44	40	40
All-Sky Solar Radiation	RadAvg	1712	1808	1597	1442	1170	1084	1138	1366	1447	1607	1679	1768	1768
	RadStd	240	174	148	109	84	97	111	114	179	174	95	124	124

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.54	-0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page